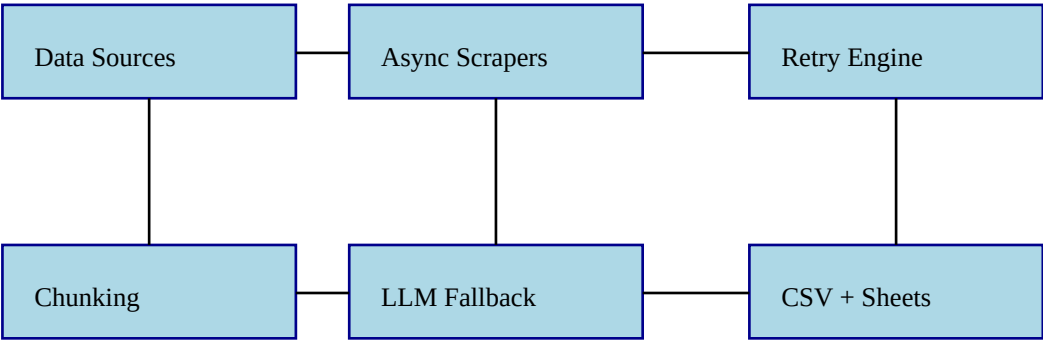


GraphOne AI Intelligence Pipeline Architecture

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Page 1: System Architecture

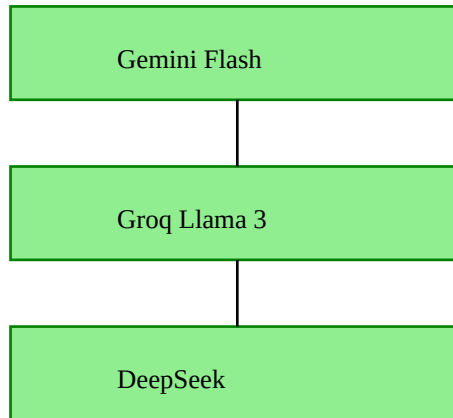


Flow: Research Papers, Jobs, News, Startups and Products are collected asynchronously, processed through retry logic and payload chunking, passed to an LLM fallback engine, deduplicated and exported to CSV/Google Sheets.

Page 2: Scalability & Reliability

Component	Strategy
Async Crawlers	asyncio + aiohttp concurrent workers
429 Handling	Exponential backoff with retry and jitter
413 Handling	Semantic chunking before LLM processing
Freshness	Jobs & News filtered to last 24 hours
Deduplication	Canonical entity mapping
Storage	CSV export for PostgreSQL / Neo4j ingestion

Page 3: LLM Orchestration



Fallback order: Gemini Flash → Groq Llama 3 → DeepSeek. Large payloads are split into semantic chunks before processing. Each chunk is summarized independently and merged into structured output.

Project Deliverables

- Startups.csv (1432 records) - Products.csv (1000+) - Research_Papers.csv (1000) - Jobs.csv (797) - News.csv (40) - Entity_Mapping_Log.csv